

Question number	Answer	Additional guidance	Mark
8(a)	- number of deaths not due to cancer calculated (1) 0.45 : 1 / 0.45 (1)	(541589 - 166800) 374 789 ACCEPT 1 : 2.25 Bald answer of 0.45 : 1 or 0.45 or 1 : 2.25 = 2 marks	(2)

	Answer	Additional guidance	Mark
8(b)(i)	An answer that includes the following points: • insertion and deletion (1)	ACCEPT chromosome or translocation IGNORE point / gene / frameshift / addition / subtraction / substitution / named disorders / missense / nonsense / inversion / duplication	(1)

Question number	Answer	Additional guidance	Mark
8(b)(ii)	65 (%)	ACCEPT any number to one decimal place in the range 58.0 to 67.0	(1)

Question number	Answer	Additional guidance	Mark
8(b)(iii)	An explanation that includes four of the following points: - substitution mutation swaps one base (in the DNA sequence / gene) (1) - this will result in a different <u>base</u> being inserted into the mRNA (during transcription) (1) - the genetic <u>code</u> is degenerate (1) - therefore codes for the same amino acid (1) - and therefore the protein will have the same {structure / function} as the R groups will be the same (1)	ACCEPT substitution only affects one base example using two sets of bases	(4)

Question number	Answer	Mark
8(c)(i)	The only correct answer is B, 0.125 <i>A is incorrect because 1 in 4 chance of having child with PKU and 1 in 2 chance of having a boy</i> <i>C is incorrect because 1 in 4 chance of having child with PKU and 1 in 2 chance of having a boy</i> <i>D is incorrect because 1 in 4 chance of having child with PKU and 1 in 2 chance of having a boy</i>	(1)

Question number	Answer	Additional guidance	Mark
*8(c)(ii)	Group 1 : Individuals 1 and 2 • must be carriers / heterozygous (S) • because they did not have PKU but some of their children did (E) Group 2 : Individuals 3 and 4 • must be homozygous (S) • because they both had PKU (E) • because both alleles need to be recessive (E) Group 3 : Individuals 5 and / or 6 • cannot tell if homozygous dominant or heterozygous / carrier (S) • because both these genotypes will give an individual without PKU (E) • we do not know anything about individual 6' family (U) • because no evidence in individuals 7, 8 or 9 of the disease (U) • so less likely that they are both carriers (U) Group 4 : Individuals 7 and / or 8 and / or 9 • cannot tell if they are homozygous dominant or heterozygous (S) • because both these genotypes will give an individual without PKU (E) • and we cannot determine genotypes of individuals 5 and 6 (U) • no offspring to give any clues (U) • but PKU is in individual 5's family (U) • so any of individuals 7, 8 or 9 could be carriers (U)	Level 1 1 mark = genotypes for one group of individuals stated OR women are XX and men are XY OR a description of what pedigree diagrams {show / can be used for} 2 marks = explanation for one group OR genotypes for two groups stated Level 2 3 marks = explanations for two groups 4 marks = explanations for three or four groups with no uncertainty Level 3 5 marks = three groups explaining the uncertainty of either group 3 or 4 6 marks = four groups explaining the uncertainty of both groups 3 and 4 NB max 2 marks if they have discussed in relation to sex linkage	(6)